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PAPER

Carbon nanotubes in benzene: internal and external solvation†

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The structure and dynamics of benzene inside and outside of single-walled carbon nanotubes (SWNTs) in the (*n,n*) armchair configuration are studied *via* molecular dynamics computer simulations. Irrespective of the nanotube diameter, benzene molecules form cylindrical solvation shell structures on the outside of the nanotubes. Their molecular planes near the SWNTs in the first external solvation shell are oriented parallel to the nanotube surface, forming a π -stacked structure between the two. By contrast, the benzene distributions in the interior of the SWNTs are found to vary markedly with the nanotube diameter. In the case of the (7,7) and (8,8) nanotubes, internal benzene forms a single-file distribution, either in a vertex-to-vertex (*n* = 7) or face-to-face (*n* = 8) orientation between two neighboring molecules. Inside a slightly wider (9,9) nanotube channel, however, a cylindrical single-shell distribution of benzene arises. A secondary solvation structure, which begins to appear inside (10,10), develops into a full structure separate from the first internal solvation shell in (12,12). The ring orientation of internal benzene is generally parallel to the nanotube wall for *n* = 9–12, while it becomes either slanted with respect to (*n* = 7), or perpendicular to (*n* = 8), the nanotube axis. The confinement inside the small nanotube pores exerts a strong influence on the dynamics of benzene. Both translational and rotational dynamics inside SWNTs are slower and more anisotropic than in liquid benzene. It is also found that reorientational dynamics of internal benzene deviate dramatically from the rotational diffusion regime and change substantially with the nanotube diameter.

1. Introduction

Recent theoretical and experimental studies have shown that a wide range of liquid properties, both structural and dynamical, can be greatly altered in a confined environment, compared to the bulk.¹ For instance, X-ray and differential scanning calorimetry (DSC) measurements² indicate that ionic liquids form crystal-like structures with elevated melting points inside carbon nanotubes. A simulation study shows that the solvation structure of ionic liquids inside the nanotubes varies dramatically with their channel diameter.³ Another intriguing and interesting example is the solid–liquid phase transition of water, for which both a continuous pathway and a discontinuous first-order phase transition appear to be possible inside of hydrophobic nanotubes and nano-slits.^{4,5} Such novel properties of the confined liquids can be profitably exploited in many diverse areas, *e.g.*, nano-science and technology. Despite this interesting potential, our understanding of the effect of

nano-confinements on materials properties is still quite limited, in particular, at the molecular level. Therefore, it is important and desirable to establish detailed theoretical pictures to understand and predict the structure and dynamics of molecular liquids, subjected to confinement, to utilize fully this potential.

As an initial step toward this goal, we study solvation of single-walled carbon nanotubes (SWNTs) in benzene *via* molecular dynamics computer simulations. Benzene is a prototypical, non-dipolar but quadrupolar, aromatic liquid with a low dielectric constant, which is immiscible with water. Because of the coupling of π -electrons of aromatic molecules and carbon nanotubes,^{6,7} benzene and its derivatives adsorb strongly to carbon nanotubes, compared to their aliphatic counterparts. The π -interaction with aromatic compounds also plays a significant role in enabling noncovalent sidewall functionalization of SWNTs, which has an important implication for developing, *e.g.*, molecular sensors and drug delivery systems through immobilization of molecules.^{8–11}

The structure and dynamics of benzene in nano-confined environments have been studied recently by experiments^{12–15} and simulations.¹⁶ According to DSC measurements,¹³ benzene confined in graphitic micropores yields crystalline states that are considerably more stable than the bulk liquid phase. In another study,¹⁵ DSC and neutron scattering measurements indicated that the melting and glass transition temperatures of

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benzene confined in silica-walled cylindrical mesoporous materials decrease and increase, respectively, as the pore diameter decreases. Spin–spin and spin–lattice relaxation times accessible *via* proton NMR show that overall dynamics of confined benzene are slower than in the bulk phase.¹⁵ Effects of confinement inside of nanopores (with average size ≥ 1.4 nm) on benzene reorientational dynamics have been studied by Raman¹² and optical Kerr effect¹⁴ spectroscopies. Regardless of the pore size, the orientational diffusion of confined benzene was found to occur on a slower time scale than in the bulk.

In this article, we investigate the structure and dynamics of benzene inside SWNT pores as well as near their exterior surfaces. We analyze in detail the influence of the pore size on the internal structure of benzene and its translational and rotational motions, and examine potential connections between confined structure and dynamics.

The outline of this paper is as follows: In Section 2, we give a brief description of the models and methods employed in this study. Solvation structure and dynamics of benzene both inside and outside of the nanotubes with differing diameters are investigated in Section 3. Concluding remarks are offered in Section 4.

2. Simulation methods

The simulation cell is comprised of a rigid SWNT in the (n,n) armchair configuration, immersed in liquid benzene consisting of 512 C₆H₆ molecules. We considered five different SWNTs, *i.e.*, $n = 7, 8, 9, 10$ and 12 . As in ref. 3, both ends of the nanotubes are terminated with hydrogen atoms, modelled by the Lennard-Jones (LJ) parameters $\varepsilon = 15.2$ K and $\sigma = 0.242$ nm and partial charge $+0.1e$ where e is the elementary charge. The C–H bond length employed for the termination is 0.109 nm. The C atoms directly bonded to the terminal H atoms have a partial charge $-0.1e$, while all others are assumed to be electrically neutral. These charge assignments, also used previously to analyze nanotube solvation in a room-temperature ionic liquid,³ are roughly based on DFT results from ref. 17. The LJ parameters employed for C atoms of the nanotubes are $\varepsilon = 43.2$ K and $\sigma = 0.340$ nm.¹⁸ During the simulations, the nanotubes were held immobile and rigid with carbon–carbon bond length $l_{CC} = 0.1415$ nm,¹⁹ and the nanotube length $h = 3.742$ nm, defined as the end-to-end, hydrogen-to-hydrogen distance. For benzene, we used an all-atom description with the LJ parameters taken from the OPLS-AA force field model²⁰ and geometry and partial charges from ref. 21.

The MD simulations were performed using the DL_POLY program (version 2.13).²² Periodic, orthorhombic boundary conditions were employed. The long-range electrostatic interactions were computed *via* the standard Ewald method,²³ resulting in essentially no truncation of these interactions. The trajectories were integrated *via* the Verlet leapfrog algorithm using a time step of 2 fs. To determine the density of each of the combined nanotube–benzene systems, we first conducted simulations in the NPT ensemble at 298 K and 1 atm for 2 ns for all five different SWNTs. Benzene molecules were allowed to move in and out of the nanotube during the

simulations. The resulting number density of benzene averaged in the NPT ensemble is $6.68, 6.67, 6.66, 6.65$ and 6.58 nm^{−3} for $n = 7, 8, 9, 10$ and 12 , respectively. The corresponding result for pure liquid benzene is 6.7 nm^{−3}. Using these densities, we simulated molecular dynamics of the nanotube–benzene systems in the canonical ensemble at 298 K. The simulations were carried out with 1 ns equilibration, followed by a 4 ns trajectory from which various ensemble averages were computed. For comparison, we also simulated pure liquid benzene with no SWNTs present to analyze dynamic properties of the bulk liquid. For additional details of the simulations, the reader is referred to the ESI.†

3. Results and discussions

3.1 Structural properties

We begin by considering solvation structures of benzene inside (“internal solvation”) and outside (“external solvation”) of the SWNTs. We employ the cylindrical coordinate system (r, ϕ, z) , where the center of the SWNT under consideration is at the origin, its axis in the z direction, with a radial distance r measured from the z -axis, and an azimuthal angle ϕ . The interior region of the (n,n) SWNT is defined as the volume V_0 of a cylinder of height h given by the nanotube length, and with a base of diameter d_0

$$d_0 = \frac{3n}{\pi} l_{CC}; \quad V_0 = \frac{\pi}{4} d_0^2 h, \quad (1)$$

where l_{CC} is the carbon–carbon bond length. A benzene molecule is considered to be inside the SWNT channel if its center of mass is located inside the cylindrical volume thus defined.

MD results for the average number of benzene molecules inside the SWNTs, N_i , hereafter referred to as the internal number, are compiled in Table 1 and the corresponding number densities, N_i/V_0 , are shown in Fig. 1a. The minimum size of SWNT that allows benzene molecules to enter the nanotube tunnel was found to be $n = 7$ with $d_0 = 0.95$ nm, under normal conditions. As n increases, so does the benzene density inside the SWNT. This is mainly due to the excluded volume effect arising from the SWNT–benzene short-range interactions. Roughly speaking, the excluded volume increases in proportion to the surface area of the SWNT interior wall, *i.e.*, linearly with d_0 , while V_0 grows quadratically with the diameter. Therefore, the net excluded volume effect decreases as d_0^{-1} with increasing V_0 . The results for the relative fluctuations in N_i measured as its standard deviation σ_N are displayed

Table 1 Number of benzene molecules inside (n,n) SWNTs^a

n	d_0/nm	N_i	σ_N
6	0.81	0	0
7	0.95	5.86	0.40
8	1.08	9.99	0.08
9	1.22	16.3	0.55
10	1.35	21.5	0.63
12	1.62	33.7	0.85

^a σ_N are the standard deviations of benzene numbers inside the nanotubes.

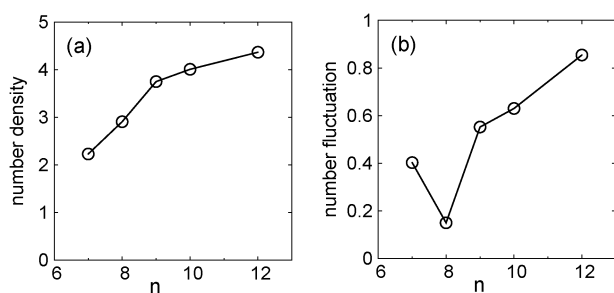


Fig. 1 (a) Number densities (units: nm^{-3}) and (b) number fluctuations of benzene molecules in the interior of (n, n) SWNTs at 298 K. The MD result for the number density of liquid benzene at 298 K and 1 atm is $n_0 = 6.7 \text{ nm}^{-3}$.

in Fig. 1b as a function of n . It is interesting that σ_N shows a non-monotonic behavior in n with a minimum at the (8,8) SWNT. Since σ_N gauges structural stability, this suggests that benzene structure inside of (8,8) is more stable than in other nanotubes (see below).

We proceed to the solvation structure of benzene. We first introduce the number of benzene molecules, $N(r, z)$, inside a cylindrical volume, extending from $z' = -z$ to $z' = z$ with a base of radius r . $N(r, z)$ is related to the local number density of benzene molecules, $\rho(r, \phi, z)$, via

$$N(r, z) = \int_0^{2\pi} \int_0^z \int_{-z}^z r' dr' d\phi' dz' \rho(r', \phi', z'), \quad (2)$$

where the center of mass is used to denote the positions of individual molecules. In terms of $N(r, z)$, the internal benzene number N_i considered above is given as

$$N_i = N(d_0/2, h/2). \quad (3)$$

We define the radial distribution of benzene molecules as²⁴

$$\begin{aligned} \xi(r) &= \frac{N(r + \Delta r, h/2) - N(r, h/2)}{2\pi r h \Delta r} = \frac{1}{2\pi n_0 r h} \frac{\partial}{\partial r} N(r, h/2) \\ &= \frac{1}{n_0} \langle \rho(r, \phi, z) \rangle_{\phi, z}, \end{aligned} \quad (4)$$

where Δr is a small increment of the radial distance, $\langle \dots \rangle_{\phi, z}$ is an average over ϕ ($0 \leq \phi < 2\pi$) and z ($-h/2 \leq z \leq h/2$) at given r . $\xi(r)$ gauges variations of the benzene density with r . In eqn (4), n_0 ($= 6.7 \text{ nm}^{-3}$) is the bulk density of benzene. We note that $\xi(r)$ is normalized in such a way that it yields the asymptotic behavior $\xi(r) \rightarrow 1$ as $r \rightarrow \infty$ just like the spherical radial distribution function, widely used in the structural analysis of bulk liquids.

The MD results for $\xi(r)$ in Fig. 2 illustrate that solvation structures of benzenes inside and outside of the SWNTs, demarcated by vertical dotted lines, differ markedly. We consider these external and internal solvation structures next.

3.1.1 External solvation. The MD results in Fig. 2 show that SWNTs are characterized by essentially the same external solvation structures regardless of the nanotube size, *viz.* diameter. For all the nanotubes studied, the radial distribution exhibited distinctive cylindrical shell-like distributions, of which the first minimum is located at 0.6 nm from the exterior wall. The first external solvation shell, *i.e.* the main structure inside the first minimum, has two peaks situated about

0.34 nm and 0.48 nm from the nanotube surface. By contrast, the second solvation shell centered around 0.8 nm from the nanotube surface shows a single-peak structure.

To further characterize solvation structure, we consider the average orientation of benzene molecules, $\Omega_2(r)$, as a function of the distance r from the SWNT axis

$$\Omega_2(r) = \langle P_2(\cos\theta) \rangle; P_2(\cos\theta) = \frac{3}{2}\cos^2\theta - \frac{1}{2}, \quad (5)$$

where θ is the angle between the normal to the benzene ring of the i th molecule and the radial direction to its center of mass, P_2 is the second-order Legendre polynomial and $\langle \dots \rangle$ means that an average is taken over only those molecules located at r . Thus, for example, $\Omega_2(r) = 1$ represents the orientation of benzene rings perpendicular to the radial direction and parallel to the SWNT surface, *viz.* $\theta = 0^\circ$, while an isotropic distribution yields $\Omega_2(r) = 0$.

The results for $\Omega_2(r)$ are shown in Fig. 3a. For clarity, plots of $\Omega_2(r)$ are presented with increasing offsets by 1.5 in the y -axis for different SWNTs. The broken line in each plot marks the $\Omega_2(r) = 0$ line. Comparison with Fig. 2 reveals that benzene molecules in the first peak of the first external solvation shell, located about 0.34 nm from the nanotube surface, mainly π -stack on its exterior wall with their rings parallel to the surface (*i.e.*, $\Omega_2(r) \approx 1$). For additional insight, we analyzed the probability distribution $P(\theta)$ of angle θ , subjected to the normalization condition $\int d\theta P(\theta) = 1$. Two sharp peaks of $P(\theta)$ near $\theta = 0^\circ$ and 180° in Fig. 3b confirm the formation of π -structure between benzene in the first peak of $\xi(r)$ and the exterior wall of SWNTs (*cf.* Fig. 4 below). It is interesting that the stacking of benzene rings on the nanotube surface occurs without explicit π -interactions between the two in our model description. Similar stacking configurations have been found between imidazolium ions and aromatic molecules in prior simulation studies.^{3,25–27}

As expected, the benzene orientation becomes isotropic ($\Omega_2(r) \approx 0$) as r increases in Fig. 3a. There are a couple of interesting features that deserve some remarks. Firstly, the transition from π -stacking to random orientation occurs over a relatively short distance ~ 0.5 nm. Thus the orientation of benzene molecules in the second solvation shell, the center of which is located only ~ 0.4 nm away from that of the first shell, is already quite isotropic. We note that rapid variations of ring orientations with an r value similar to our result here have also been observed in imidazolium-based room-temperature ionic liquids when interfaced with surfaces.^{3,28} Secondly, the transition from π -stacking to random orientations couched in terms of $\Omega_2(r)$ is non-monotonic in r . For example, the second peak of the first external solvation shell is characterized by $\Omega_2(r) \approx -0.2$, indicating that on average the benzene molecules there are tilted from the π -stacking orientation by $\sim 65^\circ$. The corresponding $P(\theta)$ in Fig. 3c shows that their orientation there is characterized by a broad distribution over the range $40^\circ \lesssim \theta \lesssim 140^\circ$.

3.1.2 Internal solvation. Now we turn to solvation structures of benzene molecules inside the nanotubes. For illustration, representative snapshots of their configurations in the interior of different nanotubes are displayed in Fig. 5,

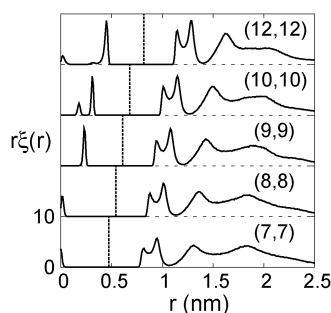


Fig. 2 Radial distributions of benzene molecules inside and outside of SWNTs at 298 K. r is the distance measured in units of nm from the nanotube axis and the dashed vertical lines denote the positions of the nanotube walls.

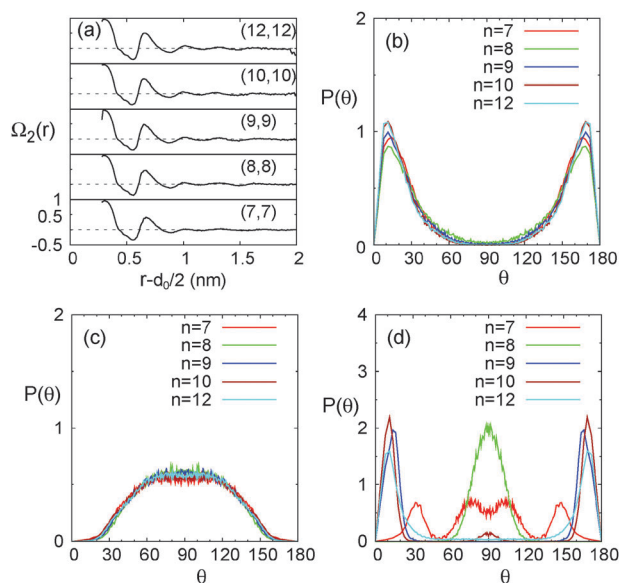


Fig. 3 Orientation of benzene molecules: (a) $\Omega_2(r)$ (eqn (5)) of benzene outside of the SWNTs; $P(\theta)$ of benzene molecules in the (b) first peak and (c) second peak of the first external solvation shell; (d) $P(\theta)$ of internal benzene. In (b)–(d), θ is the angle (units: degree) between the ring normal and the radial vector from the nanotube axis to the center of the mass of the ring, where $P(\theta)$ is normalized as $\int d\theta P(\theta) = 1$.

which reveals remarkable variations with the SWNT size. Specifically, inside the (7,7) SWNT of diameter 0.95 nm, the smallest nanotube we considered, we obtain a single file distribution of benzene, where neighboring benzene molecules are aligned in a vertex-to-vertex fashion. The single-file nature is manifested as a narrow peak in $r\xi(r)$ at $r = 0$ in Fig. 2. $P(\theta)$ in Fig. 3d indicates that the ring orientation of these benzene molecules is characterized by a rather broad distribution. With a slight increase in the nanotube diameter from 0.95 nm ($n = 7$) to 1.08 nm ($n = 8$), however, the internal distribution of benzene changes drastically. While it maintains the single file character, it forms a columnar structure with face-to-face stacking of benzene molecules with an average center-to-center separation of ~ 0.37 nm (see Fig. 5b). As a result, benzene rings inside of (8,8) are all aligned nearly in a perpendicular direction to the nanotube axis in total contrast to the (7,7)

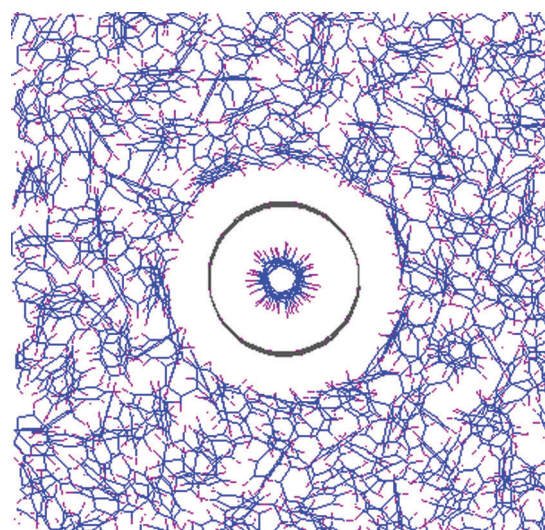


Fig. 4 Snapshot of a benzene configuration in the presence of a (8,8) SWNT.

case. This is well exposed by its $P(\theta)$ centered at $\theta = 90^\circ$ in Fig. 3d. It is interesting to note that similar orientation is also obtained when only one benzene molecule is present inside the (8,8) nanotube tunnel.²⁹ The extremely low σ_N value in Table 1 and Fig. 1b suggests that the internal distribution in Fig. 5b is a well-ordered structure, reminiscent of the discotic columnar liquid crystal phase. The exceedingly small mobility of benzene along the SWNT axis inside (8,8) compared to other nanotubes lends further support to this view (see Table 2 and Fig. 8 below). Nevertheless, its radial distribution looks very similar to that of (7,7) because the center of mass is employed to represent the benzene position. As the SWNT size further increases to $d_0 = 1.22$ nm, corresponding to the $n = 9$ case, the internal benzene structure undergoes another dramatic change; benzene molecules stack instead on the interior wall of the nanotube (Fig. 3d and 5c) and develop a shell-like distribution. Their ring centers are located about 0.25 nm from the nanotube axis.

In the interior of (10,10), with diameter 1.33 nm, a second solvation structure appears around $r = 0.22$ nm inside of the first internal solvation shell situated at $r \approx 0.35$ nm. The small separation between the two suggests that their benzene molecules are intercalated. We thus expect that the two solvation structures are interconnected and not completely separated in three dimensions. By contrast, in the case of (12,12) with $d_0 = 1.62$ nm, its larger channel size allows the formation of a distinct second internal solvation structure near $r = 0$. Two peaks of $P(\theta)$ at $\theta = 10^\circ$ and 170° in Fig. 3d indicate that benzene molecules in the first internal shell mainly form a π -stack on the (12,12) nanotube wall.

We also consider density profiles of benzene molecules along the nanotube axial direction, *i.e.* the z -axis. Following ref. 3, we introduce the dimensionless z -distribution $\zeta(z; b, a)$

$$\zeta(z; b, a) = \frac{1}{\pi n_0 (b^2 - a^2)} \int_a^b \int_0^{2\pi} r' dr' d\phi' \rho(r', \phi', z), \quad (6)$$

which gauges the density averaged over the cylindrical shell

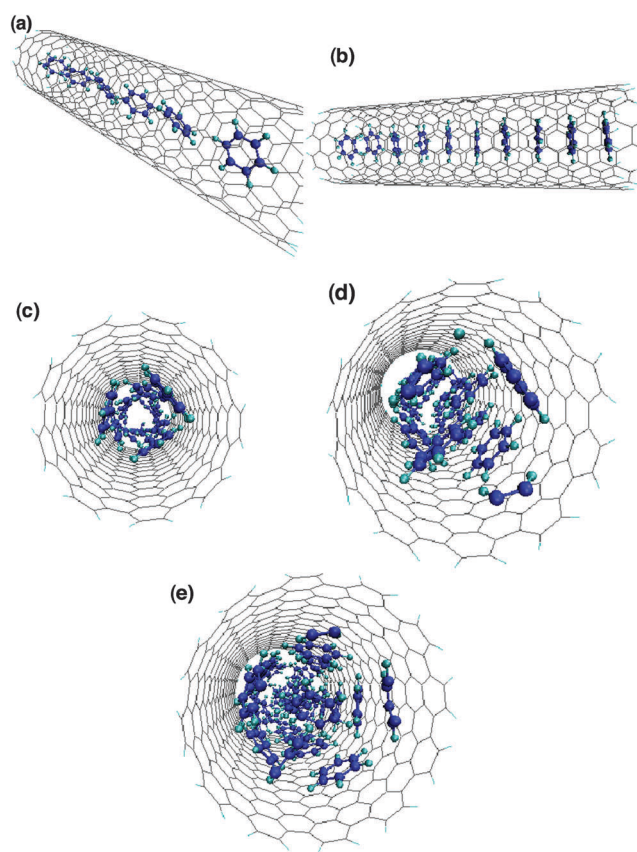


Fig. 5 Snapshots of benzene molecules in the interior of (a) (7,7), (b) (8,8), (c) (9,9), (d) (10,10), and (e) (12,12) SWNTs.

Table 2 Translational diffusion coefficients (units: fick) for benzene inside nanotubes

<i>n</i>	$D_{\parallel}$	$D_{\perp}$	$\mathcal{R}_{\text{tr}} (= D_{\perp}/D_{\parallel})$
7	0.12	~ 0	~ 0
8	0.0012	~ 0	~ 0
9	0.090	0.022	0.24
10	0.020	0.011	0.55
12	0.020	0.016	0.8
Bulk ^a	1.7	1.7	1

^a In liquid benzene.

region between $r = a$ and b ($a < b$) at given z . The z -distribution and $N(r, z)$ in eqn (2) are related via

$$\pi n_0 r^2 [\zeta(z; r, 0) + \zeta(-z; r, 0)] = 2 \int_0^r \int_0^{2\pi} r' dr' d\phi' \rho(r', \phi', z) \quad (7)$$

$$= \frac{\partial}{\partial z} N(r, z).$$

We notice in Fig. 6 that the z -distribution of benzene in the interior of SWNTs varies quite significantly with the size of the tunnel. Its amplitude for $n = 7$ and 8 oscillates between 0 and 0.6–0.7 with nearly a constant interval along z throughout the entire nanotube channel. This regular oscillatory behavior is a clear sign that on average, individual benzene molecules (or more accurately their center of mass) occupy well-defined locations inside the (7,7) and (8,8) nanotubes, and thus form a

more crystal-like structure than the liquid. It should be noted that the number of peaks in the z -distribution matches exactly the corresponding internal benzene number in Table 1. For larger nanotubes ($n = 9$ –12) in Fig. 6c–e, the oscillatory character of the z -distribution becomes greatly reduced and its magnitude becomes nearly independent of z . This suggests that internal solvation structure lacks crystal-like ordering. However, the presence of regular oscillations of a tiny amplitude for (9,9) suggests that its internal benzene structure is ordered to a certain degree. As for the z -distribution outside of SWNTs, with the exception of $n = 12$, benzene in the first external solvation shell does not show any distinctive structures.

Before we turn to dynamic properties, we consider briefly whether the distributions of internal benzene show any chirality. According to several prior studies, solvents like water^{30–34} and ionic liquids³ can form chiral distributions inside nanotube micropores of a particular size. To examine this possibility quantitatively, we first label the internal benzene molecules in the descending order of their z -coordinate values and introduce a relative angle ψ_i between two neighboring molecules, i and $i + 1$, defined as³

$$\psi_i \equiv \phi_{i+1} - \phi_i \quad (-180^\circ < \psi_i < 180^\circ). \quad (8)$$

In eqn (8), the positions (r_i, ϕ_i, z_i) of benzene molecules satisfy $z_1 > z_2 > z_3 > \dots$. If the radial vector of i is rotated by ψ_i , it will be in the direction of that of $i + 1$. If the sign of ψ_i does not change for the group of molecules under consideration, their distribution is in general chiral.³ We have analyzed ψ_i for the $n = 9, 10$ and 12 cases by employing 100 configurations for each SWNT, sampled every 40 ps from its 4 ns MD trajectory. Only in the (9,9) channel, does the internal distribution of benzene exhibit some degree of chirality as shown in Fig. 7. Nonetheless, sign changes in ψ_i there indicates that the extent of chirality of the benzene distribution inside (9,9) is considerably weaker than that of the confined imidazolium-based ionic liquid studied in ref. 3. Other nanotubes do not yield any chirality in their internal benzene distributions, and their results are not shown.

3.2 Dynamic properties

3.2.1 Translational dynamics. To understand translational dynamics of benzene confined to the inside of SWNTs, we investigate mean-square displacements (MSD)

$$\text{MSD}(t) = \frac{1}{N} \sum_i \langle |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \rangle, \quad (9)$$

where $\mathbf{r}_i$ is the center of mass of the i th benzene, $\langle \dots \rangle$ is an equilibrium ensemble average, and N is the number of benzene molecules considered in the sum. In the normal diffusion regime in three dimensions, which usually occurs in the time scale well beyond the velocity correlation time, the self-diffusion coefficient D is related to MSD via $\text{MSD}(t) = 6Dt$. In the case of single-file diffusion in narrow pores, however, MSD increases as $\sqrt{t}$ in the long-time limit.^{35–38} In the present study, we examine the dynamic behaviors of benzene molecules inside the nanotube pore and in the first external solvation shell, focusing on the influence of nanotube size. Since our system is not isotropic, we consider translational motions of

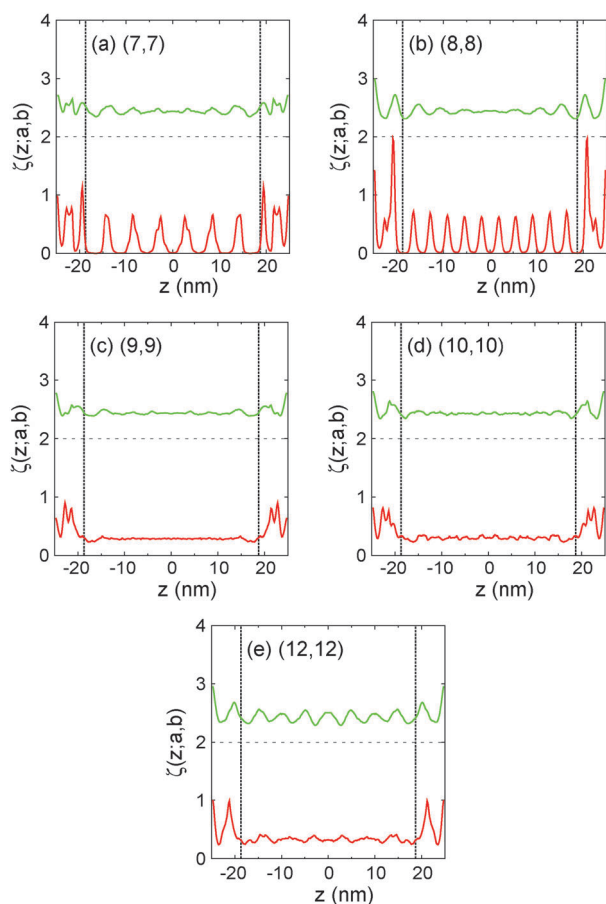


Fig. 6 z -Distributions $\zeta(z; b, a)$ of benzene molecules in the first external solvation shell ($a = d_0/2$, $b = r_{\min}$) and in the interior ($a = 0$, $b = d_0/2$) of SWNTs. $r_{\min}$ is the location of the minimum in $\xi(r)$ between the first and second external solvation shells. For clarity, $\zeta(z; b, a)$ of the first external solvation shell (green) is shifted upwards by 2 with respect to that of the internal solvation shell (red) in each panel. The dashed vertical lines denote the positions of terminal H atoms of the nanotubes along z .

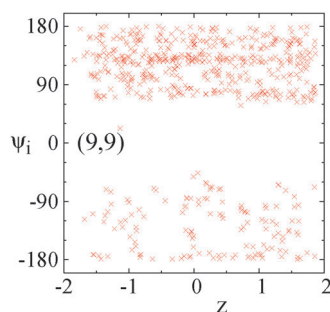


Fig. 7 Rotational angle ψ_i for benzene molecules inside a (9,9) nanotube.

benzene along the nanotube axis, *i.e.* the z -direction, and in the plane normal to the axis separately, and their respective diffusion coefficients, $D_{\parallel}$ and $D_{\perp}$. For an isotropic system, $3D = D_{\parallel} + 2D_{\perp}$ with $D_{\parallel} = D_{\perp}$.

The MD results for $\text{MSD}(t)$ and diffusion coefficients are shown in Fig. 8 and 9. For convenience, the results for $D_{\parallel}$ and

$D_{\perp}$ for internal benzene are also compiled in Table 2. We note that benzene molecules located initially inside the nanotube pore or in the first external solvation shell do not remain in the same region indefinitely. Thus in each time window used to calculate $|\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2$ in eqn (9), we considered only those benzene molecules that stay in the same region throughout the entire time window.³⁹ Therefore the number of benzene molecules that contribute to $\text{MSD}(t)$ varies not only with the initial configuration of time windows, but also with their duration.

Fig. 8 shows that after initial ballistic motions that last for ~ 5 ps, benzene translational motions tend to become diffusive both inside and outside of the nanotubes. Comparison with the bulk liquid shows clearly that the presence of SWNTs exerts a strong influence on benzene dynamics, especially in the region close to the nanotubes. Specifically, the translational diffusivity near the nanotube is considerably lower than in the bulk and varies significantly with the nanotube pore size. As expected, their translational dynamics are anisotropic, *i.e.*, $\mathcal{R}_{\text{tr}}(\equiv D_{\perp}/D_{\parallel}) \neq 1$.

The $D_{\parallel}$ and $D_{\perp}$ values for benzene in the first external solvation shell decrease monotonically as the SWNT diameter increases. By contrast, those in the interior of nanotubes exhibit a non-monotonic behavior. Inside the (10,10) and (12,12) nanotubes, benzene diffusion coefficients are $D_{\parallel} = 0.02$ fick and $D_{\perp} = 0.011$ – 0.016 fick ($1 \text{ fick} = 10^{-9} \text{ m}^2 \text{ s}^{-1}$), which are smaller than the bulk value of 1.7 fick by two orders of magnitude. As we decrease the SWNT size to $n = 9$, $D_{\parallel}$ and $D_{\perp}$ increase to 0.09 and 0.022 fick, respectively. For (9,9), (10,10) and (12,12), $D_{\perp}$ arises mainly from benzene motions in the azimuthal direction, *i.e.* changes in their ϕ -coordinate values on the xy -plane, rather than those along the radial direction. This can be inferred easily from very narrow internal

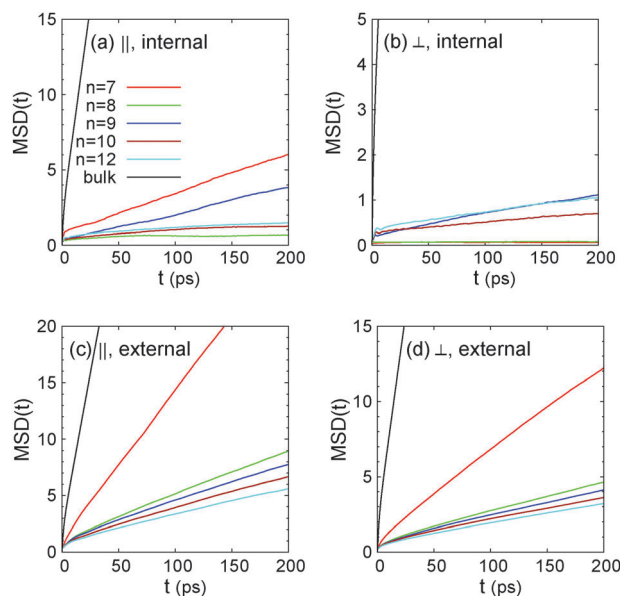


Fig. 8 Mean square displacements $\text{MSD}(t)$ along the direction of the nanotube axis ($\parallel$) and in the plane normal to the axis ($\perp$) in the interior and in the first external solvation shell of (n, n) SWNTs. For comparison, $\text{MSD}(t)$ in bulk benzene is also presented. $\text{MSD}(t)$ is measured in units of 10^{-2} nm^2 .

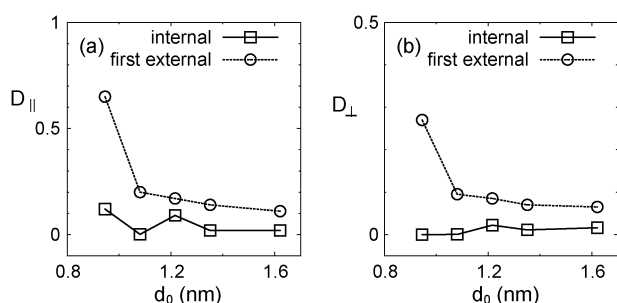


Fig. 9 Translational diffusion coefficients of benzene molecules, measured in fick, (a) along the axial z -direction and (b) in the plane normal to the axis in the interior (—) and in the first external solvation shell (---) of (n, n) SWNTs at 298 K. The $D_{||} (= D_{\perp})$ value in the bulk liquid benzene is 1.7 fick, according to our simulation.

peaks in Fig. 2. This also implies that internal benzene molecules generally maintain their π -stacking structure as they undergo translational motions. If we continue to lower the nanotube diameter to 1.08 nm, corresponding to (8,8) SWNT, both $D_{||}$ and $D_{\perp}$ drop dramatically. $D_{||}$ decreases by a factor of 80 to 0.0012 fick, compared to the $n = 9$ case, and $D_{\perp}$ essentially vanishes. The latter aspect can be understood easily as the small nanotube diameter does not allow much room for internal diffusion in the radial or azimuthal direction. The marked reduction in $D_{||}$ on the other hand is attributed to the formation of the columnar structure of π -stacked benzene molecules inside (8,8) as mentioned in Section 3.1.2 above. As we further reduce the SWNT size to $n = 7$, $D_{||}$ of internal benzene rises drastically by two orders magnitude to 0.12 fick. This abrupt increase is ascribed to the breaking of the ordered columnar structure and decrease in the internal benzene density. The $D_{\perp}$ value remains close to zero due to tight confinement in the radial direction. We also point out that $\mathcal{R}_{tr}$ decreases, *i.e.* translational anisotropy increases with decreasing SWNT size because motions of internal benzene along x and y become progressively more restricted.

Another noteworthy result is that despite the quasi-one-dimensional nature of SWNT channels, we did not find any evidence of single-file diffusion^{35–38} for internal benzene. This state of affairs is analogous to that of water in narrow carbon nanotubes, studied recently.⁴⁰ One likely reason for the absence of single-file diffusion is the short length of the nanotubes considered in our simulations.

3.2.2 Reorientational dynamics. Now we turn to rotational dynamics of benzene inside the nanotubes. As per usual, we consider reorientational time correlation functions $C_{\Omega}^{(l)}(t)$ defined as

$$C_{\Omega}^{(l)}(t) \equiv \langle P_l[\cos\theta^{\Omega}(t)] \rangle; \tau_{\Omega}^{(l)} = \int_0^{\infty} C_{\Omega}^{(l)}(t) dt, \quad (10)$$

where P_l is the l th Legendre polynomial, $\theta^{\Omega}(t)$ is the angle between the molecular orientation $\hat{\Omega}$ of interest at time t and its initial direction, and $\tau_{\Omega}^{(l)}$ is the correlation time associated with $C_{\Omega}^{(l)}(t)$. We consider two different reorientational motions for benzene, *i.e.* reorientations of the C_6 -axis normal to the ring plane and in-plane C_2 -axes perpendicular to the C_6 -axis. The corresponding orientations will be denoted as “nml” and “in”, respectively. Analogous to $\mathcal{R}_{tr}$, we introduce the

rotational anisotropy factor as $\mathcal{R}_{rot}^{(l)} = \tau_{nml}^{(l)} / \tau_{in}^{(l)}$. We also define spinning and tumbling motions of benzene as rotations about its C_6 - and C_2 -axes, respectively.^{41,42} Tumbling is solely responsible for the relaxation of $C_{nml}^{(l)}(t)$, *i.e.* reorientation of the C_6 -axis, while both spinning and tumbling contribute to reorientations of the C_2 -axes, $C_{in}^{(l)}(t)$.

The MD results for benzene inside the nanotubes are presented in Fig. 10 and Table 3. For comparison, the results in pure benzene are also given there. We notice that the reorientational dynamics of internal benzene vary substantially with the SWNT diameter and deviate markedly from those in liquid benzene. With the exception of the $n = 7$ and 8 cases, the relaxation of $C_{in}^{(l)}(t)$ tends to be faster than that of $C_{nml}^{(l)}(t)$, *i.e.* $\tau_{in}^{(l)} < \tau_{nml}^{(l)}$ with $\mathcal{R}_{rot}^{(l)} > 1$. This is in reasonable accord with several prior studies of bulk^{41–44} and confined^{12,14,15} benzene. The $\mathcal{R}_{rot}^{(l)}$ results in Table 3 show that the rotational anisotropy is enhanced markedly inside the narrow SWNT pores, compared to bulk benzene. Analogous to translational motions, reorientations of both C_6 - and C_2 -axes become decelerated dramatically inside the SWNTs. For the bulk phase, MD yields $\tau_{in}^{(1)} = 5.0$, $\tau_{in}^{(2)} = 2.1$, $\tau_{nml}^{(1)} = 8.9$ and $\tau_{nml}^{(2)} = 3.0$ ps with $\mathcal{R}_{rot}^{(l)} = 1.8$ and 1.4 for $l = 1$ and $l = 2$, respectively. The fact that $\tau_{\Omega}^{(1)} \approx 3\tau_{\Omega}^{(2)}$ for both $\Omega = \text{“in”}$ and “nml” suggests that to a good approximation, reorientational dynamics of C_2 - and C_6 -axes fall in the rotational diffusion regime in the bulk. By following ref. 42, we evaluated

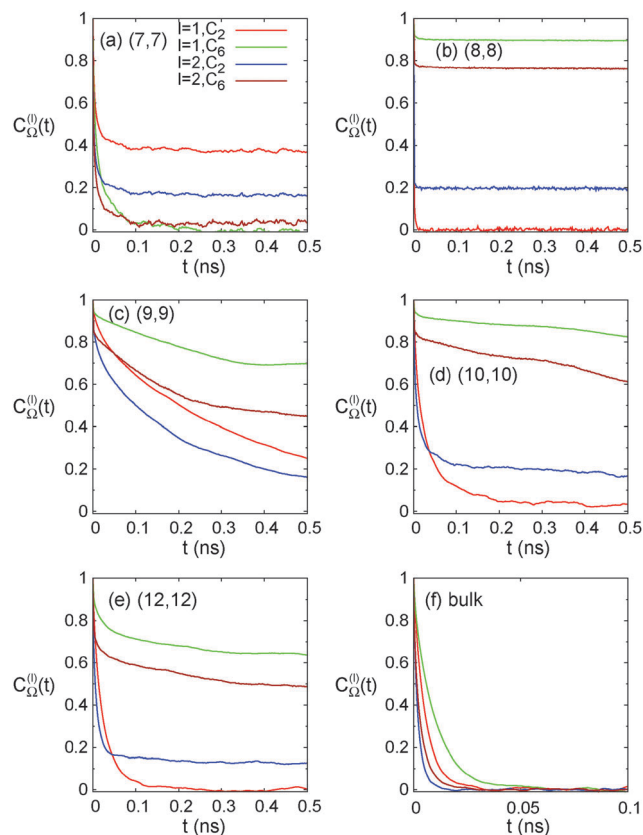


Fig. 10 Reorientational time correlation functions $C_{\Omega}^{(l)}(t)$ of the C_2 - and C_6 -axes of benzene inside (n, n) SWNTs at 298 K. Only the $l = 1$ and 2 components are shown. For reference, $C_{\Omega}^{(l)}(t)$ in the bulk liquid benzene are presented in (f).

Table 3 MD results for reorientational correlation times $\tau_{\Omega}^{(l)}$ of benzene^a

<i>n</i>	$\tau_{\text{in}}^{(1)}$	$\tau_{\text{nml}}^{(1)}$	$\mathcal{R}_{\text{rot}}^{(1)}$	$\tau_{\text{in}}^{(2)}$	$\tau_{\text{nml}}^{(2)}$	$\mathcal{R}_{\text{rot}}^{(2)}$
7	3100	20	0.0065	530	36	0.068
8	2.3	1.6×10^5	7.0×10^4	∞	3.1×10^4	0.0
9	340	1200	3.5	220	470	2.1
10	60	4600	77	290	1400	4.8
12	26	2200	85	230	950	4.1
Bulk ^b	5.0	8.9	1.8	2.1	3.0	1.4

^a Units: ps. ^b In liquid benzene.

rotational diffusion coefficients from $C_{\Omega}^{(2)}(t)$. The results are $D_{\text{sp}}^{\text{rot}} = 110 \times 10^9 \text{ s}^{-1}$ for spinning and $D_{\text{tm}}^{\text{rot}} = 56 \times 10^9 \text{ s}^{-1}$ for tumbling in the bulk.

In contrast to bulk benzene, $C_{\text{in}}^{(1)}(t)$ and $C_{\text{in}}^{(2)}(t)$ inside the (12,12) SWNT intersect at $t \approx 45$ ps (Fig. 10e), disclosing a major departure from rotational diffusion. As a result, $\tau_{\text{in}}^{(2)}$ is much greater than $\tau_{\text{in}}^{(1)}$. For a better understanding, we analyzed the Green function $G_{\Omega}(\theta; t)$ associated with reorientational dynamics.⁴⁵

$$G_{\Omega}(\theta; t) = \langle \delta(\theta^{\text{rot}}(t) - \theta) \rangle = \sum_l \frac{2l+1}{2} P_l(\cos \theta) \sin \theta C_{\Omega}^{(l)}(t); \quad (11)$$

$$C_{\Omega}^{(l)}(t) = \int_0^{\pi} P_l(\cos \theta) G_{\Omega}(\theta; t) d\theta.$$

$G_{\text{in}}(\theta; t)$ in the interior of (12,12) in Fig. 11e shows a striking deviation from that in the bulk liquid in Fig. 11f. Conditional probabilities at $\theta \approx 60^\circ$, 120° and 170° grow rapidly with time, indicating that reorientations of the C_2 -axes through rotations by multiples of 60° are favored. Also of interest is the fact that $G_{\text{in}}(\theta; t)$ for $t \gtrsim 100$ ps is nearly invariant under 60° rotations of θ , *viz.* $\theta \rightarrow \theta + 60^\circ$. This is possible if rotations do not involve tumbling, *i.e.* reorientations occur *via* spinning, so that the directions of three C_2 -axes before and after a 60° rotation coincide. A rather narrow distribution of $G_{\text{nml}}(\theta; t)$, with a negligible probability for $\theta \gtrsim 75^\circ$ in the entire t range in Fig. 12e, confirms a very limited role played by tumbling in the benzene reorientational dynamics inside (12,12). We ascribe this to a closely-packed internal solvation structure of benzene, which does not allow much room for tumbling other than that arising from benzene motions along the azimuthal direction with basic π -stacking structure maintained (see Section 3.2.1 above). The (10,10) case is similar to (12,12) although reorientational dynamics in the former are slower than in the latter.

$C_{\text{in}}^{(l)}(t)$ inside of the (9,9) micropore differ substantially from those of (10,10) and (12,12). The $l = 1$ and 2 components of $C_{\text{in}}^{(l)}(t)$ in the former do not intersect, at least during the 1 ns we have investigated, and their biphasic relaxation behavior is much weaker than in (10,10) and (12,12). While their $G_{\text{in}}(\theta; t)$ share some general features, the conditional probability inside (9,9) no longer exhibits invariance under $\theta \rightarrow \theta + 60^\circ$, in contrast to (10,10) and (12,12). This means that tumbling plays a more prominent role in the former, as indicated by its wider $G_{\text{nml}}(\theta; t)$, than in the latter. The single-shell solvation structure, with a significant void in the middle (*cf.* Fig. 5c) as well as increase in benzene translations along the azimuthal

direction (*cf.* Fig. 8b), is believed to be responsible for the enhancement in tumbling in (9,9).

The reorientational dynamics of benzene inside of (8,8) are quite unique in that their Green functions do not evolve with t after rapid initial decay in the first few ps. As a result, $C_{\Omega}^{(l)}(t)$ does not show any appreciable relaxation after ~ 2 ps. The fact that $G_{\text{in}}(\theta; t)$ is nearly constant in the entire θ range for $t \gtrsim 2$ ps, while $G_{\text{nml}}(\theta; t)$ is characterized by a narrow distribution centered around $\theta = 15^\circ$, indicates that the benzene molecules undergo essentially free spinning rotations without tumbling. Analogous to (10,10) and (12,12) above, this is attributed to closely-packed structure of benzene—in this case, tight columnar π -stacking (Fig. 4 and 5b above)—which permits only spinning. We parenthetically note that as t increases, $C_{\Omega}^{(2)}(t)$ of a free symmetric top approaches 0.25,^{46–48} which compares well with our result 0.2 for $C_{\text{in}}^{(2)}(t)$.

The internal benzene of the (7,7) nanotube shows interesting reorientational behaviors, not shared by that of other SWNTs. Specifically, (7,7) is the only case where reorientations of the C_6 -axis of internal benzene are faster than those of the C_2 -axes. In addition, its $G_{\text{nml}}(\theta; t)$ becomes widened much more than other nanotubes as t increases. Thus full 180° tumbling of a benzene molecule (*viz.* a 180° rotation of its C_6 -axis) is allowed inside of (7,7), whereas C_6 -axis reorientations are restricted essentially to $\theta \lesssim 90^\circ$ for $n = 8$ –12. We believe that the low density (Fig. 1) and loose structure of internal benzene in the former are mainly responsible for the large-amplitude tumbling motions.

Though not presented here, we have analyzed angular velocity correlations of internal benzene. The interested reader is referred to the ESI.†

4. Concluding remarks

In this article, we have studied the structure and dynamics of benzene inside and outside of single-walled carbon nanotubes in the (*n,n*) armchair configuration using molecular dynamics computer simulations. It was found that benzene molecules form smeared-out cylindrical shell structures outside of the nanotubes. For all the nanotubes we studied, the first external solvation shell shows a double-peak structure. Benzene molecules in the first peak there tend to form a π -stack on the exterior walls of the nanotubes, whereas those in the second peak show a broad distribution in their ring orientations. As the distance from the nanotube increases, the orientation of benzene rings become isotropic rapidly, so that the ring orientation of the second external solvation shell is nearly random with respect to the radial direction regardless of the nanotube size.

We found that the distribution of benzene inside the nanotubes varies rather dramatically with their diameter. To be specific, solvation structures inside the (*n,n*) SWNT channel are characterized by a single-file distribution either in the vertex-to-vertex ($n = 7$) or face-to-face ($n = 8$) orientation of neighboring benzene molecules, by a cylindrical single-shell distribution ($n = 9$) and by double-shell structures ($n = 10$ and 12). Furthermore, benzene molecules in the first internal solvation shell primarily π -stack on the interior nanotube wall for $n = 9$, 10 and 12. The highly-ordered and

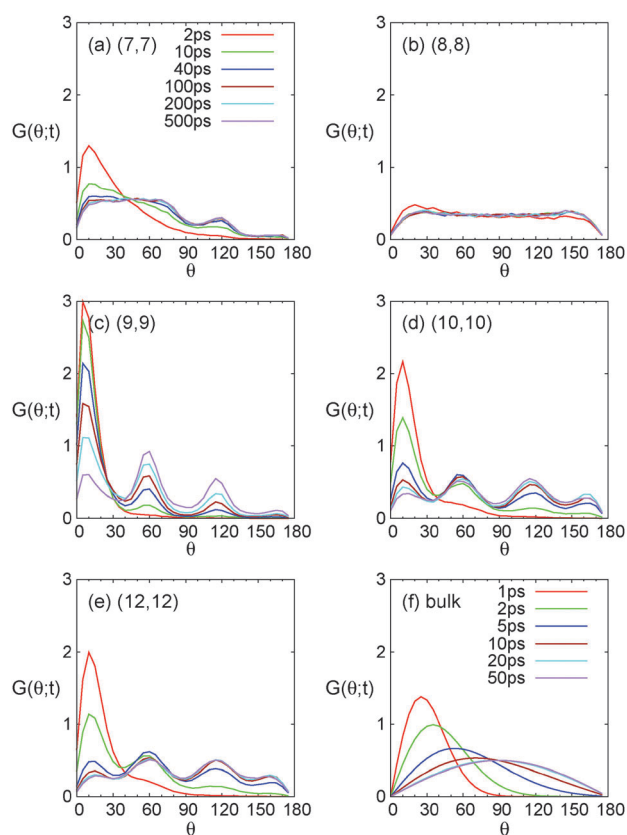


Fig. 11 MD results for the Green function $G_{in}(\theta; t)$ associated with reorientations of C_2 -axes of benzene inside (n, n) nanotubes. $G_{in}(\theta; t)$ at different t are denoted by different colors as indicated in panel (a). For comparison, the results in liquid benzene are displayed in (f).

π -stacked columnar benzene distribution which results inside of the (8,8) SWNT alludes to the formation of a stable mesophase, totally different from the bulk. Extremely slow translational dynamics of benzene in the columnar structure offers another strong piece of evidence for the high stability of this phase.

The confinement to the inside of small nanotubes exerts a strong influence on benzene dynamics. Both translational and reorientational dynamics are much slower in the nanotube channel than in the bulk. Their anisotropy becomes enhanced significantly in the former and varies substantially with the nanotube diameter. For all cases we studied, translational dynamics of internal benzene did not develop characteristics of single-file diffusion, presumably due to the short length of the nanotubes employed in the simulations. Reorientational dynamics of benzene in the nanotube channels show a pronounced departure from rotational diffusion. Also with the exception of the (7,7) SWNT case, their tumbling characteristics become significantly reduced, compared to spinning. Our analysis indicates that modulations of reorientational dynamics inside the nanotubes are closely related to the internal benzene structure, especially the π -stacking structure.

There are several different directions in which the present study can be extended in the future. Firstly, it would be desirable to study the thermal behavior of the present system and analyze the stability of the differing internal structures

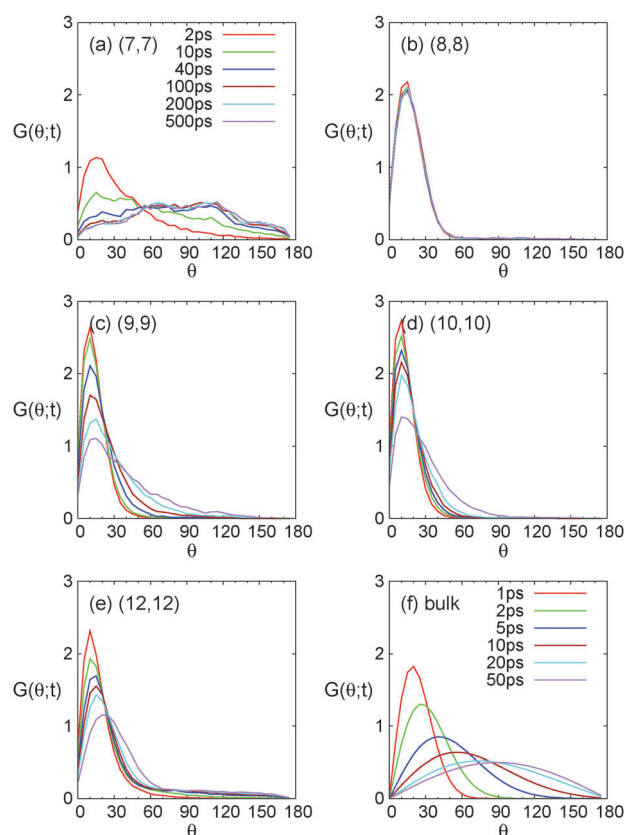


Fig. 12 MD results for $G_{nm1}(\theta; t)$ for C_6 -axis reorientations of benzene inside SWNTs. The notations and time scales are the same as in Fig. 11.

found here. Secondly, it would be interesting to study other aromatic systems but with reduced symmetry, such as toluene and anisole. Mixtures are another interesting system that is worthy of theoretical attention. Finally, it would be worthwhile to investigate different pore geometries, e.g. nano-sized slits and spheres, and examine their influence on internal benzene structures and dynamics.

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References

- For example, see: *Dynamics in Small Confining Systems IV*, ed. J. M. Drake, J. Klafter, R. Kopelman, Materials Research Society, Warrendale, PA, 1999, vol. 543, p. 372.
- S. Chen, G. Wu, M. Sha and S. Huang, *J. Am. Chem. Soc.*, 2007, **129**, 2416–2417.
- Y. Shim and H. J. Kim, *ACS Nano*, 2009, **3**, 1693–1702.
- K. Koga, G. T. Gao, H. Tanaka and X. C. Zeng, *Nature*, 2001, **412**, 802–805.
- S. Han, M. Y. Choi and H. E. Stanley, *Nat. Phys.*, 2010, **6**, 685–689.

- 6 C. Wei, D. Lin and Z. Dongqiang, *Environ. Sci. Technol.*, 2007, **41**, 8295–8300.
- 7 L. M. Woods, S. C. Bădescu and T. L. Reinecke, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2007, **75**, 155415.
- 8 E. S. Snow, F. K. Perkins, E. J. Houser, S. C. Bădescu and T. L. Reinecke, *Science*, 2005, **307**, 1942–1945.
- 9 M. Penza, F. Antolini and M. V. Antisari, *Sens. Actuators B*, 2004, **100**, 47–59.
- 10 H. Gao and Y. Kong, *Annu. Rev. Mater. Res.*, 2004, **34**, 123–150.
- 11 R. J. Chen, Y. Zhang, D. Wang and H. J. Dai, *J. Am. Chem. Soc.*, 2001, **123**, 3838–3839.
- 12 J. Yi and J. Jonas, *J. Phys. Chem.*, 1996, **100**, 16789–16793.
- 13 A. Watanabe, T. Iiyama and K. Kaneko, *Chem. Phys. Lett.*, 1999, **305**, 71–74.
- 14 X. Zhu, R. A. Farrer and J. T. Fourkas, *J. Phys. Chem. B*, 2005, **109**, 12724–12730.
- 15 Y. Xia, G. Dosseh and C. Alba-Simionesco, *J. Phys. Chem. B*, 2006, **110**, 19735–19744.
- 16 R. G. Winkler and R. Hentschke, *J. Chem. Phys.*, 1993, **99**, 5405–5417.
- 17 D. Lu, Y. Li, U. Ravaioli and K. Schulten, *J. Phys. Chem. B*, 2005, **109**, 11461–11467.
- 18 G. Hummer, J. C. Rasaiah and J. P. Noworyta, *Nature*, 2001, **414**, 188–190.
- 19 T. W. Odom, J. L. Huang, P. Kim and C. M. Lieber, *Nature*, 1998, **391**, 62–64.
- 20 J. Pranata, S. G. Wierschke and W. L. Jorgensen, *J. Am. Chem. Soc.*, 1991, **113**, 2810–2819.
- 21 P. V. Kumar and M. Maroncelli, *J. Chem. Phys.*, 1995, **103**, 3038–3060.
- 22 T. R. Forester and W. Smith, *DL POLY 2.13 user manual*, CCLRC, Daresbury Laboratory, Daresbury, Warrington, U.K., 2001.
- 23 D. M. Heyes, *J. Chem. Phys.*, 1981, **74**, 1924–1929.
- 24 Y. Shim and H. J. Kim, *ACS Nano*, 2010, **4**, 2345–2355.
- 25 C. G. Hanke, A. Johansson, J. B. Harper and R. M. Lynden-Bell, *Chem. Phys. Lett.*, 2003, **374**, 85–90.
- 26 Y. Shim, M. Y. Choi and H. J. Kim, *J. Chem. Phys.*, 2005, **122**, 044510.
- 27 S. Maolin, Z. Fuchun, W. Guozhong, F. Haiping, W. Chunlei, C. Shimou, Z. Yi and H. Jun, *J. Chem. Phys.*, 2008, **128**, 134504.
- 28 C. Pinilla, M. G. Del Pópolo, R. M. Lynden-Bell and J. Kohanoff, *J. Phys. Chem. B*, 2005, **109**, 17922–17927.
- 29 S. Y. Bhide and S. Yashonath, *J. Phys. Chem. B*, 2000, **104**, 11977–11986.
- 30 E. B. Barros, A. Jorio, G. G. Samsonidze, R. B. Capaz, A. G. Souza, F. J. Mendes, G. Dresselhaus and M. S. Dresselhaus, *Phys. Rep.*, 2006, **431**, 261–302.
- 31 K. H. Lee and S. B. Sinnott, *J. Phys. Chem. B*, 2004, **108**, 9861–9870.
- 32 Z. Mao and S. B. Sinnott, *J. Phys. Chem. B*, 2000, **104**, 4618–4624.
- 33 Y. Liu, Q. Wang, T. Wu and L. Zhang, *J. Chem. Phys.*, 2005, **123**, 234701.
- 34 Y. Liu and Q. Wang, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2005, **72**, 085420.
- 35 T. E. Harris, *J. Appl. Probab.*, 1965, **2**, 323–338.
- 36 D. G. Levitt, *Phys. Rev. A*, 1973, **8**, 3050–3054.
- 37 J. Kärgner, *Phys. Rev. A*, 1992, **45**, 4173–4174.
- 38 M. Kollmann, *Phys. Rev. Lett.*, 2003, **90**, 180602.
- 39 For example, see: B. Mukherjee, P. K. Maiti, C. Dasgupta and A. K. Sood, *J. Chem. Phys.*, 2007, **126**, 124704.
- 40 A. Striolo, *Nano Lett.*, 2006, **6**, 633–639.
- 41 R. Witt, L. Sturz and A. Dölle, *J. Phys. Chem. A*, 2000, **104**, 5716–5725.
- 42 R. J. Berry, D. Duan and M. Schwartz, *J. Phys. Chem. A*, 2005, **109**, 8637–8641.
- 43 M. A. McCool, A. F. Collings and L. A. Woolf, *J. Chem. Soc., Faraday Trans. I*, 1972, **68**, 1489–1497.
- 44 P. Linse and S. Engström, *Chem. Phys. Lett.*, 1985, **115**, 95–100.
- 45 Y. Shim, D. Jeong, M. Y. Choi and H. J. Kim, *J. Chem. Phys.*, 2006, **125**, 061102.
- 46 R. G. Gordon, *J. Chem. Phys.*, 1966, **44**, 1830–1836.
- 47 W. Steele, *Adv. Chem. Phys.*, 1976, **34**, 1–104.
- 48 A. G. Pierre and W. Steele, *Phys. Rev.*, 1969, **184**, 172–186.